INFO 5602 INFORMATION VISUALIZATION

Individual Project Presentation

(Real World Chart Evaluation)

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Chart 1: Context

- **Why:** why was this chart created?
    - Predict how the Omicron variant of COVID-19 will spread
- **Who:** who is this chart for?
    - The public and subscribers to the Times
- **What:** what did the creators want the target audience to know/do?
    - Warn people about the rise in exposure risk to COVID-19
- **How:** what data source did the chart creators use?
    - Data sourced by Columbia University (later made available on the NYTimes GitHub)

https://www.nytimes.com/2022/01/06/opinion/omicron-covid-us.html

Chart 1: Before

- Unconventional spiral time axis
  - Very hypnotic
- Unclear how many covid cases are confirmed each week
- While legend defines a scale, comparing continuous area creates more cognitive load

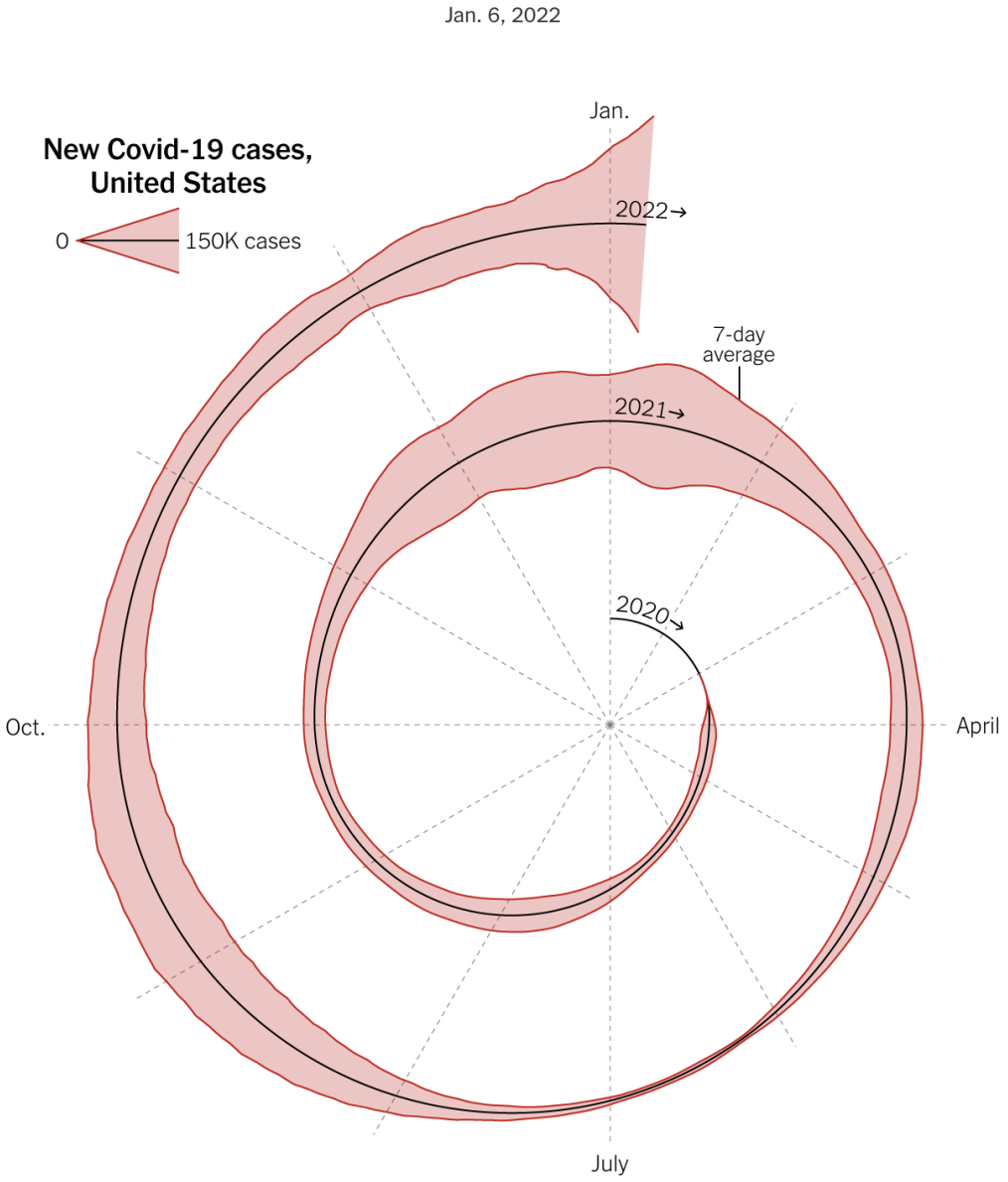


Chart 1: After

- Time runs along a linear x-axis
  - Linear time is back
- Y-axis clearly defines the amount of covid cases
  - No need to guess how many cases of covid were confirmed
- Used color and annotations to mark significant variants and when they appeared

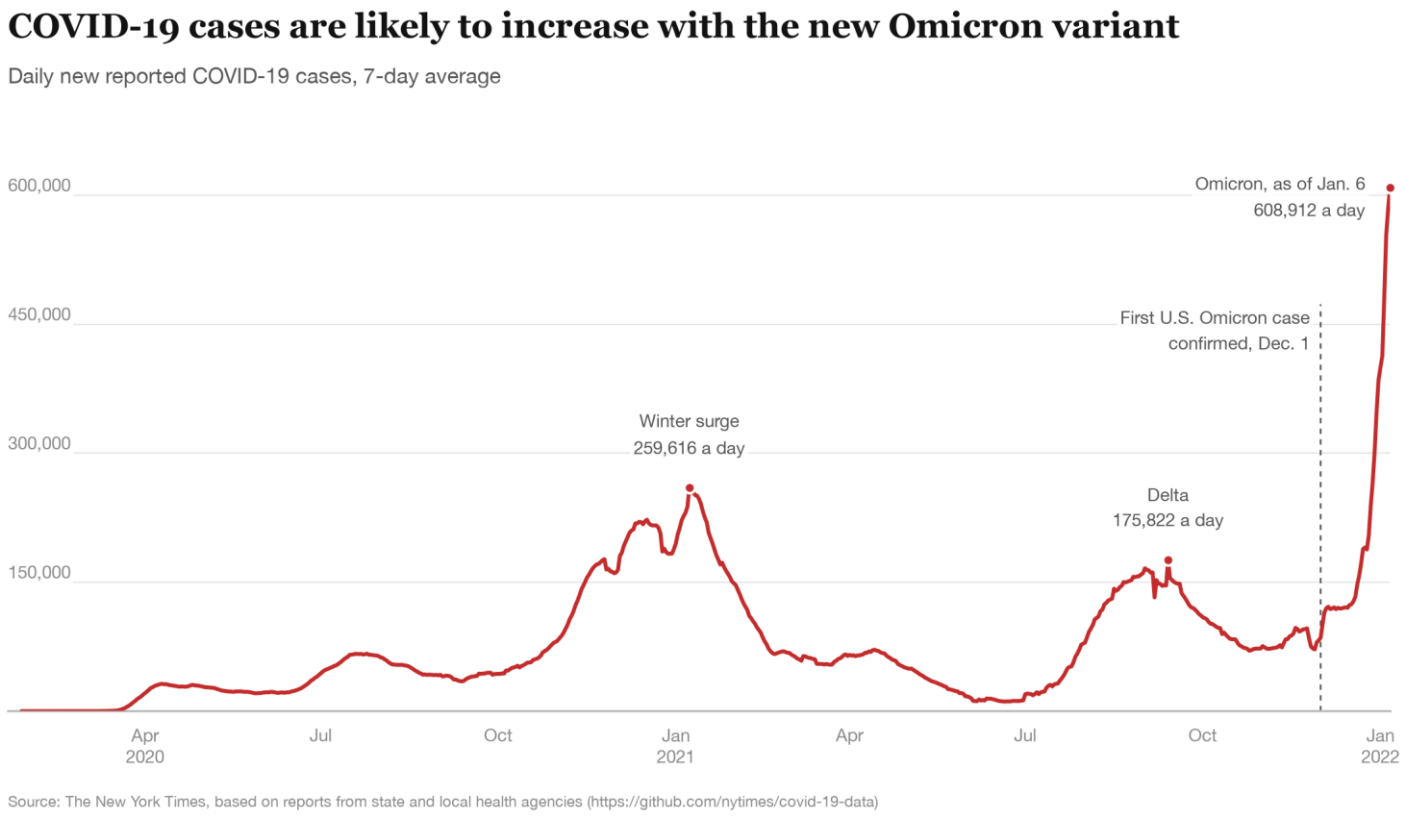
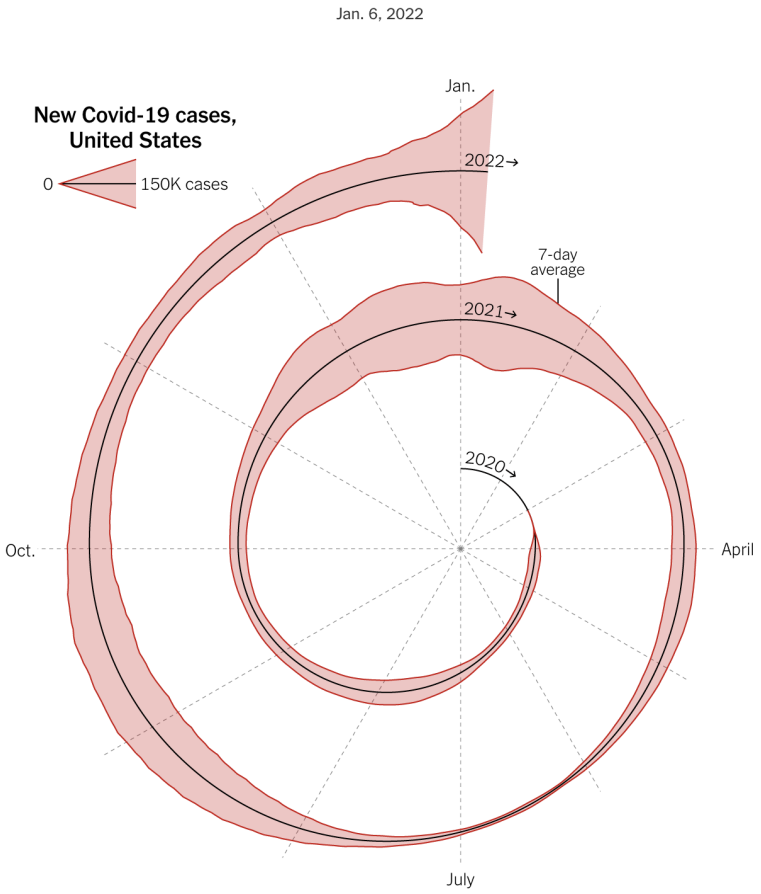


Chart 1: Before vs After



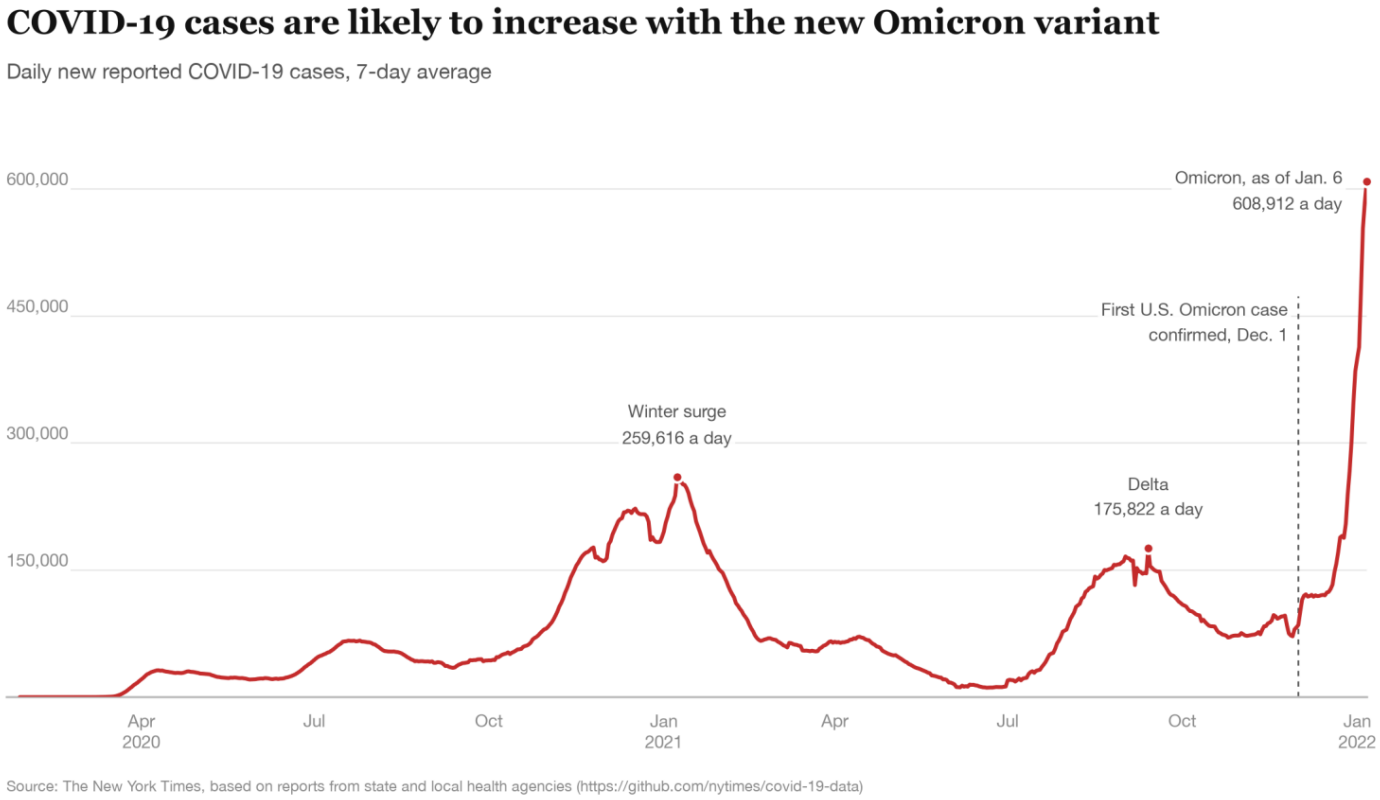


Chart 2: Context

- **Why:** why was this chart created?
    - Visualize/summarize how Americans view other G20 countries in terms of favorability
- **Who:** who is this chart for?
    - The public, government representatives, etc.
- **What:** what did the creators want the target audience to know/do?
    - Inform the audience of American opinions about other countries
- **How:** what data source did the chart creators use?
    - Pew Research Center surveyed ~5,000 adults from September 14 to 20, 2026.

https://www.pewresearch.org/global/2026/10/16/how-do-americans-feel-about-g20-countries/

Chart 2: Before

- Lines comparing political party intensify cognitive load
- Dots on the thermometer represent more than one country
  - Makes the comparison between parties difficult
- Tries to do two things at once
  - Favorite country per party
  - Comparing countries across parties
- This could have been an email
  - (This could have been a bar chart)

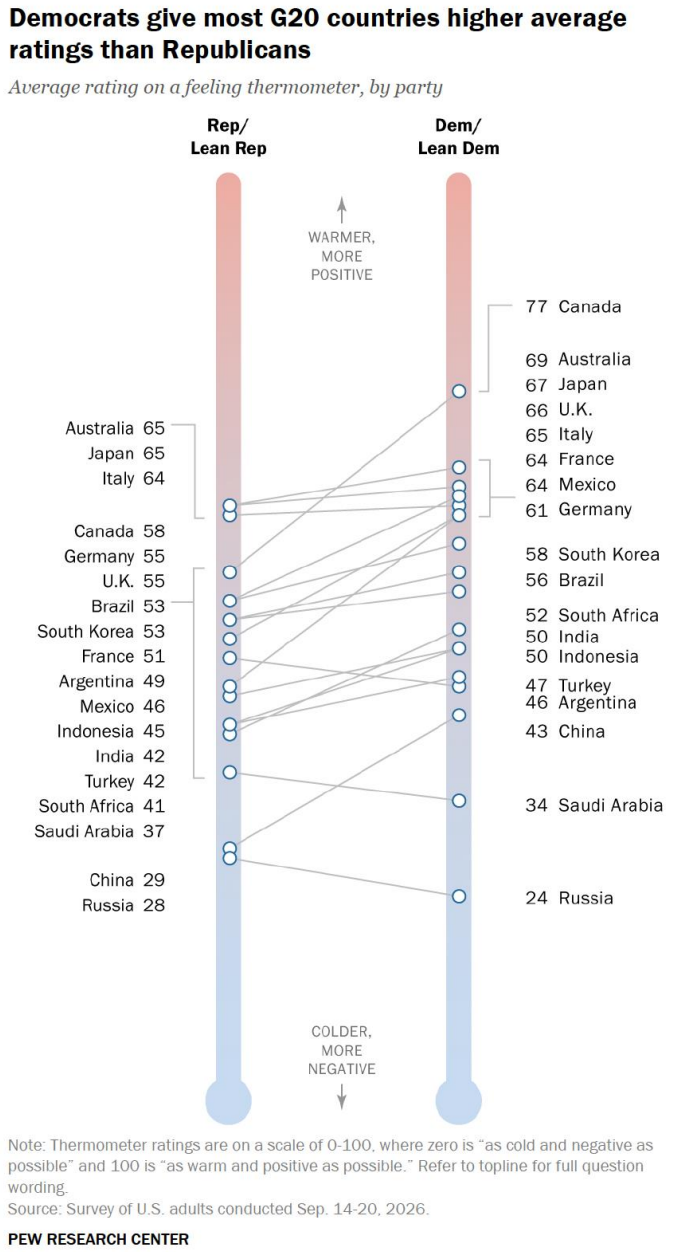


Chart 2: After

- Reduced cognitive load by grouping by country for easier comparison
- Sorted countries by overall average of both parties
  - Shows which countries are most favorable, regardless of party
- Added a party-gap column to show how much favorability differs between parties

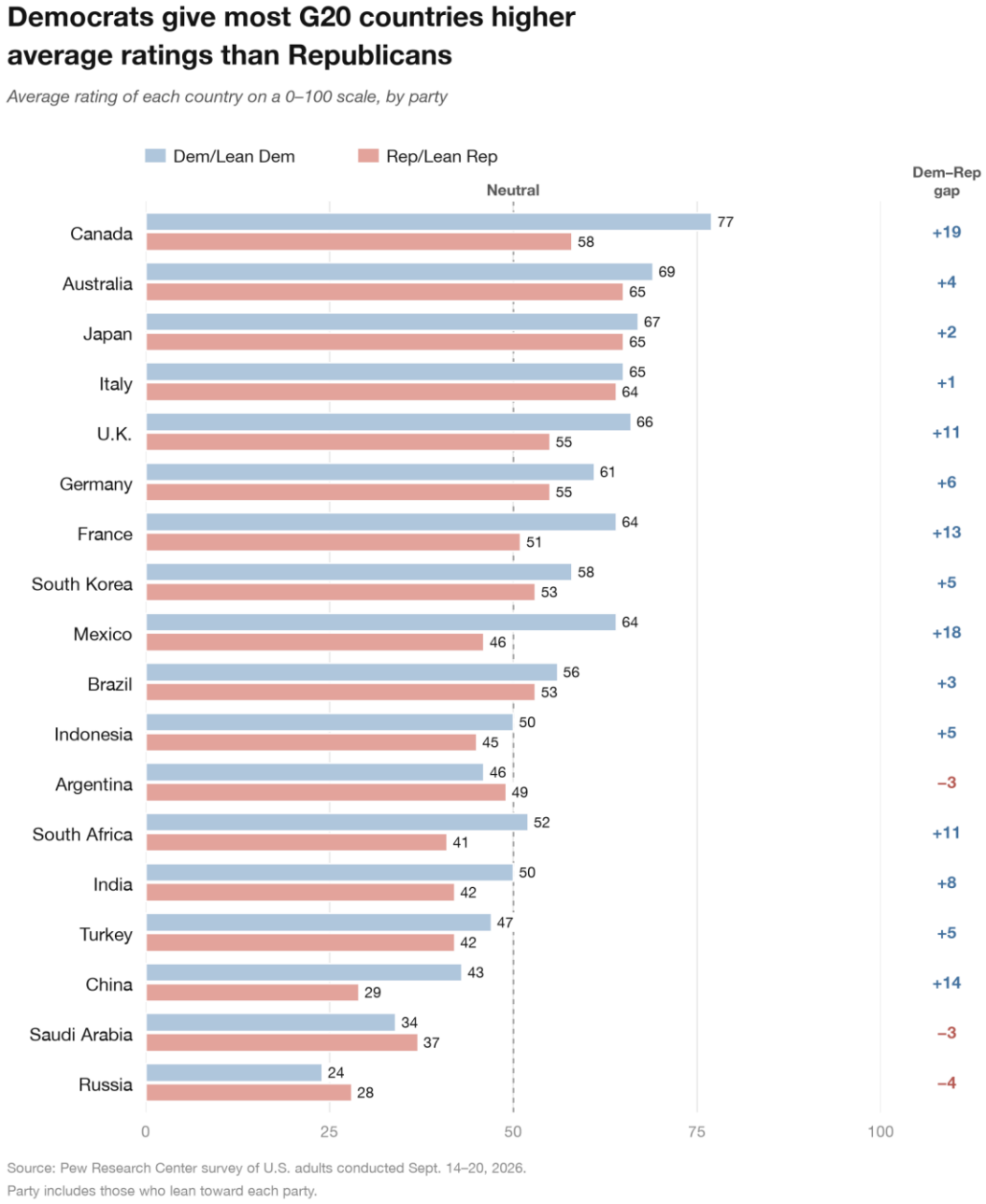
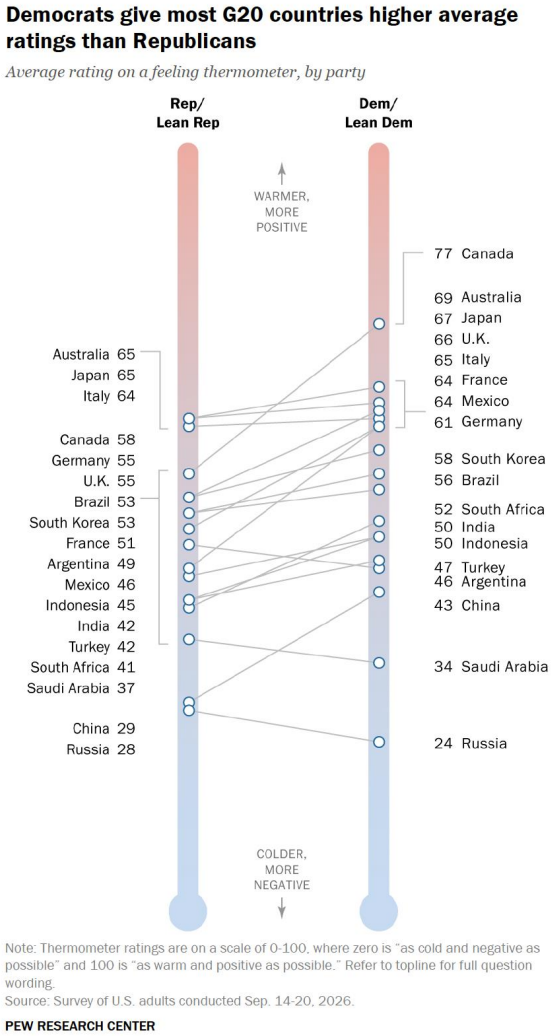


Chart 2: Before vs After



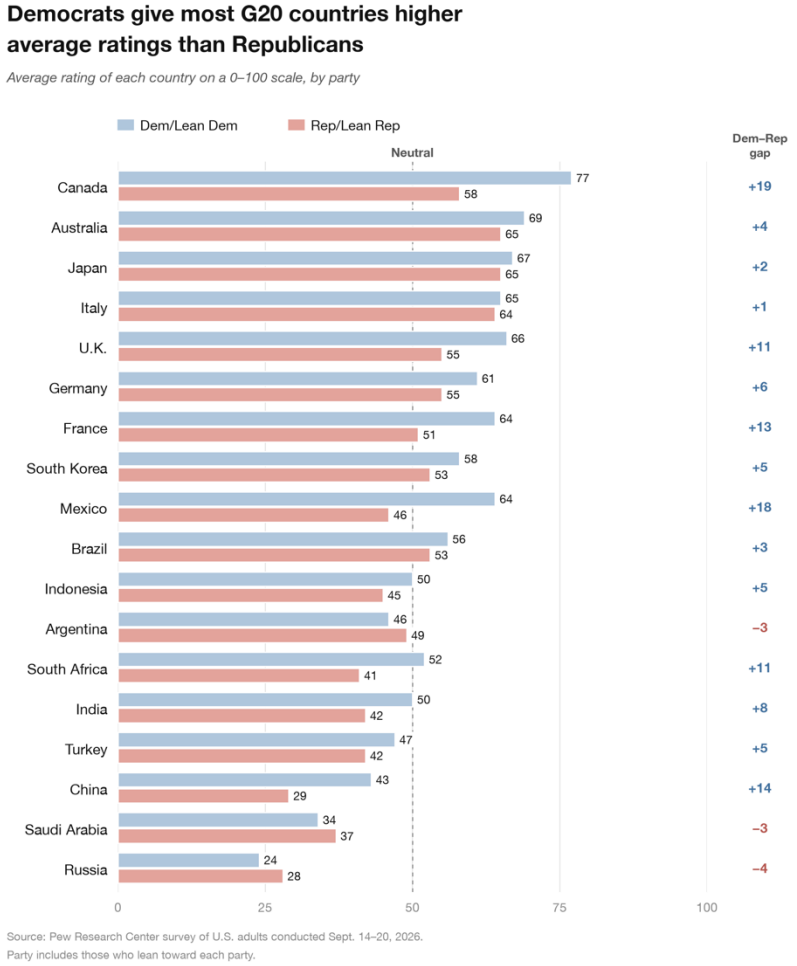


Chart 3: Context

- **Why:** why was this chart created?
    - Emphasize how the U.S.-Israeli war against Iran might impact oil reserves.
- **Who:** who is this chart for?
    - The public, other media outlets, etc.
- **What:** what did the creators want the target audience to know/do?
    - Inform the audience of how the U.S.-Israeli war against Iran can impact our daily lives.
- **How:** what data source did the chart creators use?
    - Kpler: real-time data analytics platform tracking physical commodity flows.

https://www.nytimes.com/2026/06/12/business/energy-environment/iran-war-oil-reserves.html

Chart 3: Before

- Individual year lines are redundant in the shaded year range
  - Especially if you give up halfway through labeling
- Clarifying annotations are only making the chart more confusing
- Trying to do two things at once (again)
  - Range of oil in storage reserves
  - Weekly levels per year
- Labels for other years are scattered throughout the chart

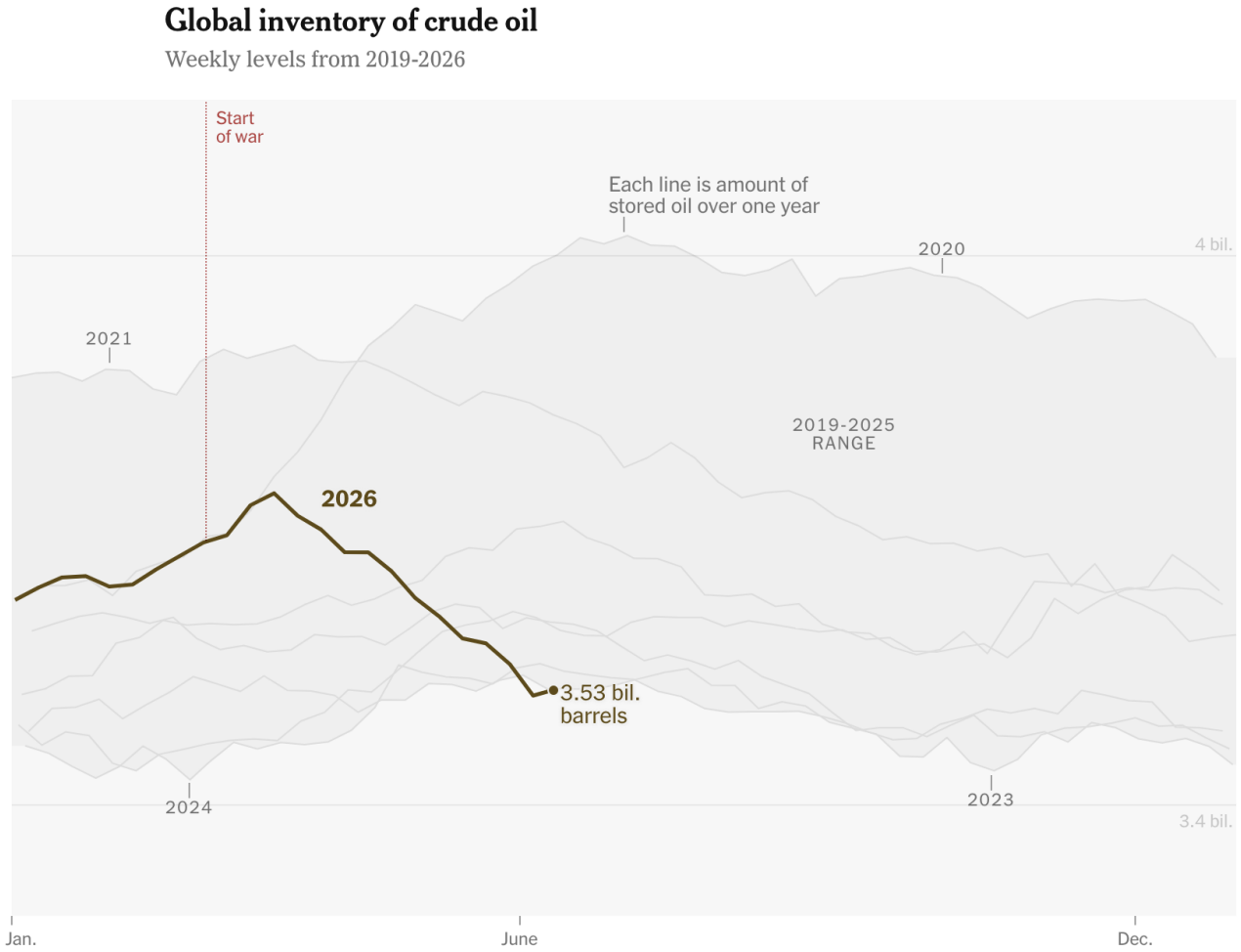
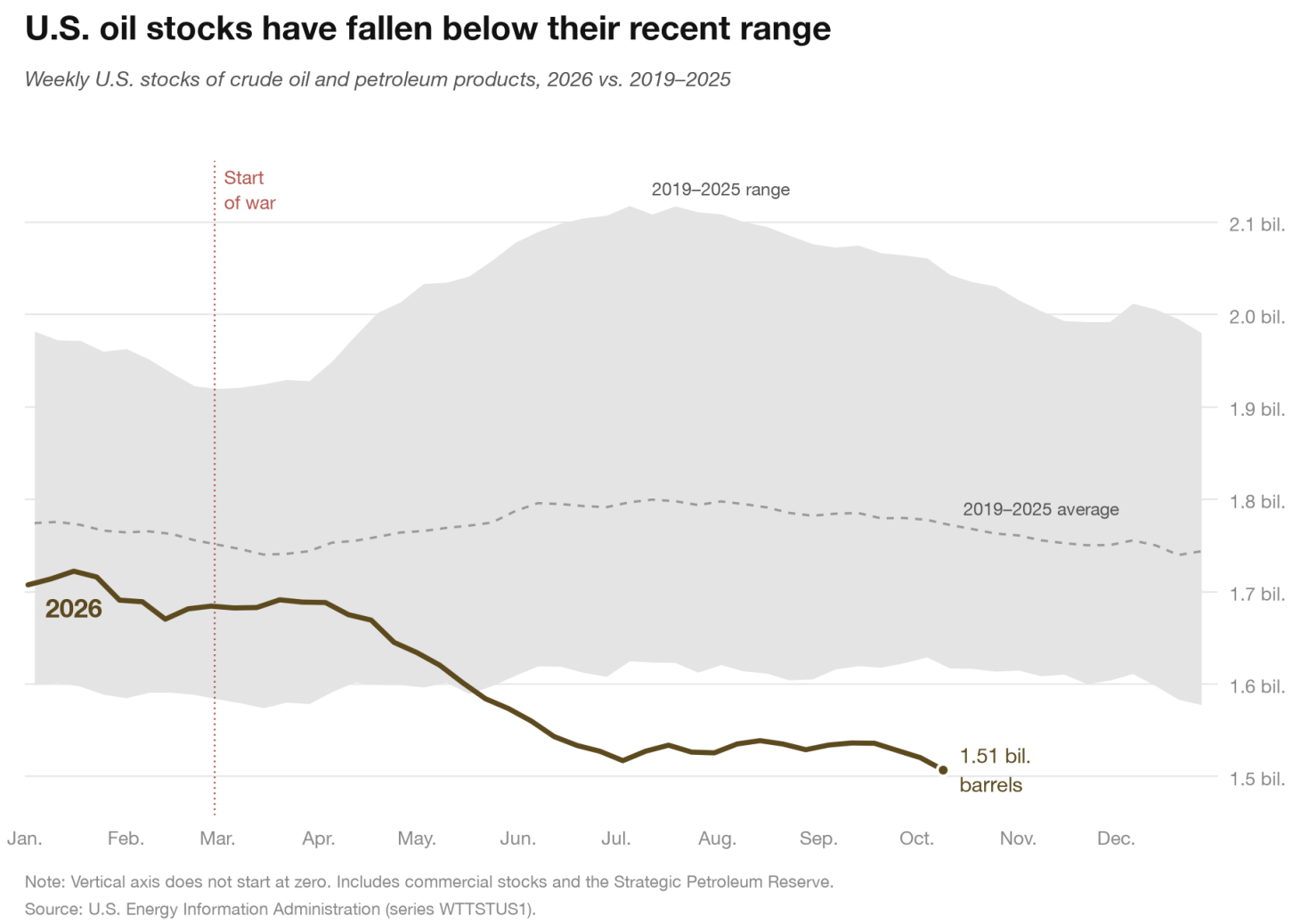


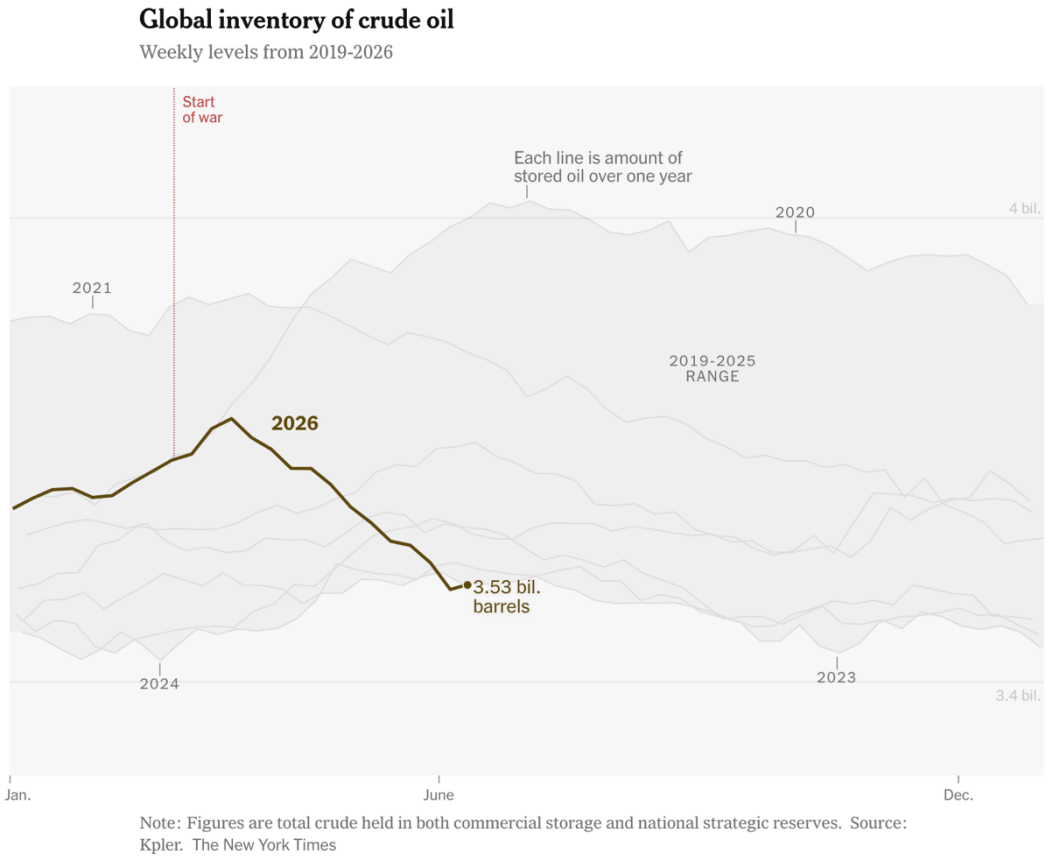
Chart 3: After

- Removed individual year lines
  - Range is all that is needed for comparison
- Only two clear annotations
  - 2019-2025 range
  - 2019-2025 average
- Added average inventory line for 2019-2025
  - Reduces cognitive load by minimizing data to filter through

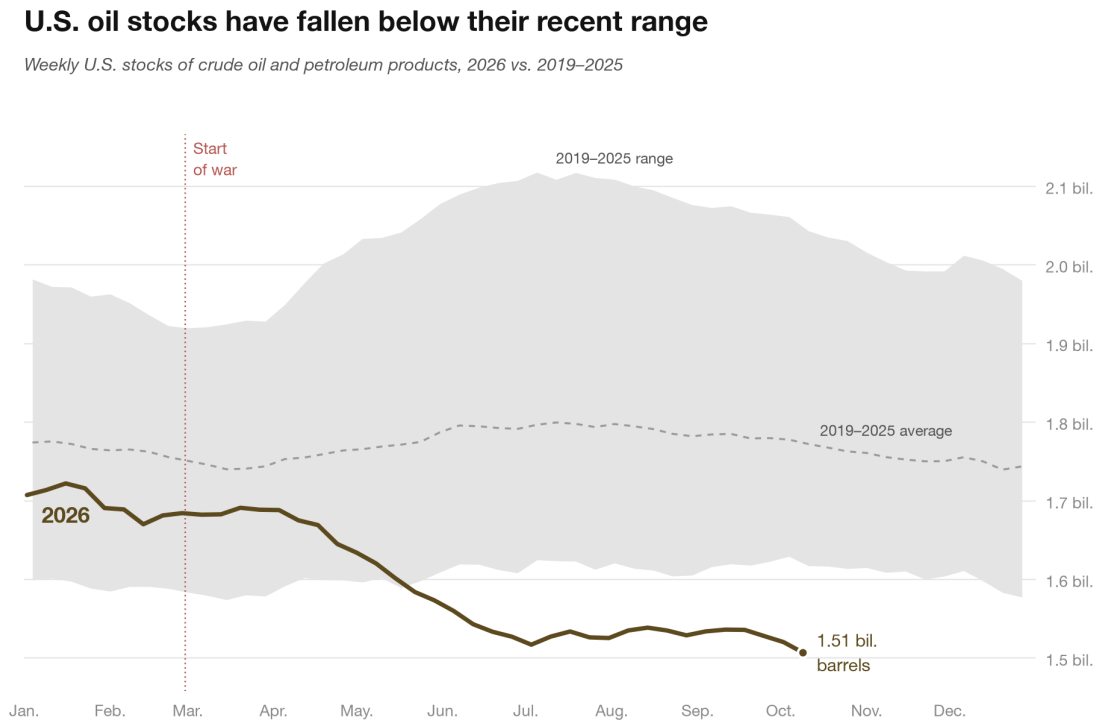


Note: I did not have access to the global inventory data, so I changed it to the U.S. national inventory to reduce fabricatio n of data.

Chart 3: Before vs After



Note: Figures are total crude held in both commercial storage and national strategic reserves. Source: Kpler. The New York Times



Note: Vertical axis does not start at zero. Includes commercial stocks and the Strategic Petroleum Reserve.
Source: U.S. Energy Information Administration (series WTTSTUS1).

Questions?